

Case Study Rubric: Fake News Text Classification

Submission Format

Submit a final report, notebook, or slide deck that walks through your reproduction of the fake news text classification case study. Your submission should include your reproduced workflow, key outputs, model evaluation results, and written interpretation of the model's strengths and limitations.

General Description

In this case study, you will reproduce a fake news detection project using text classification. The original project uses article text and headlines from a fake/real news dataset to train a machine learning model that predicts whether an article is real or fake. The goal is not to build a perfect misinformation detector. The goal is to understand the project workflow, reproduce the main steps, and explain how text data can be transformed into a machine learning prediction.

Why am I doing this?

This case study gives you practice reproducing and explaining an existing data science project. In real data science work, you often need to understand someone else's code, recreate their results, evaluate whether their methods make sense, and communicate findings clearly to others. By completing this case study, you will learn how raw article text becomes numerical data, how a logistic regression classifier can make predictions, and why results must be interpreted carefully when working with misinformation.

What am I going to do?

You will clone or download the case study repository, load the fake/real news dataset, prepare the text data, reproduce the exploratory analysis, convert the article text into TF-IDF features, train the logistic regression classifier, and evaluate the model's performance. After reproducing the workflow, you will create your own final report, notebook, or slide deck that explains the process, results, limitations, and possible improvements.

Tips for Success

- Do not overcomplicate the model. A clear explanation of the original workflow is more valuable than adding advanced methods without explaining them.
- Focus on interpretation, not just accuracy. Explain what the model can and cannot conclude.
- Use the repository materials carefully. Your goal is to reproduce and understand the provided project, not create a completely unrelated analysis.

How will I know I have succeeded?

You will meet expectations if your final product clearly explains how the fake news classifier works from beginning to end. A successful submission should show that you understand the dataset, preprocessing steps, TF-IDF features, logistic regression model, evaluation results, and limitations of automated fake news detection.

Spec Category	Specification Details
Setup and Data	Environment: Clone/download the case study repository and open the project in Python, Jupyter, Colab, or VS Code. Dataset: Load the fake/real news dataset and confirm that the article text/headline fields and real/fake labels are present.
Data Preparation	Cleaning: Prepare the text data by following the project workflow, including handling missing values, combining text fields if needed, and preparing the labels. Split: Create or verify the training/testing split used for model evaluation.
Exploratory Data Analysis	Visuals: Reproduce key EDA outputs from the case study, such as class balance, article length, common words, or word frequency. Findings: Explain what the EDA shows about the real and fake news articles.
Text Features	TF-IDF: Convert the text data into numerical features using TF-IDF. Explanation: Briefly explain why text must be converted into numbers before it can be used in a machine learning model.
Modeling	Classifier: Train the logistic regression fake news classifier from the case study workflow. Purpose: Explain why logistic regression is a reasonable baseline model for predicting real vs. fake news.
Evaluation	Metrics: Report the model's evaluation results using accuracy, precision, recall, F1-score, and/or a confusion matrix. Interpretation: Explain what the results mean in the context of fake news detection.
Discussion	Limitations: Discuss at least two limitations of the model or dataset, such as dataset bias, overfitting, changing news language, or the fact that the model learns text patterns rather than truth. Improvements: Suggest at least one way the project could be improved or extended.
Final Report	Product: Create a clear final report, notebook, or slide deck that walks through the reproduced case study from start to finish.